TRACE: Context-Probed Region Allocation for Test-Time Adaptation in Open-Vocabulary Semantic Segmentation

Anonymous submission

Abstract

Test-time adaptation learns from a segmenter's own predictions, but initial confidence need not determine how those predictions should share an update. TRACE measures an additional response: it preserves each predicted region, changes its surroundings, and reads support for the original query at the original locations. These detached supports allocate regional entropy within each source image; region averaging removes an explicit area multiplier and image normalization fixes each active image's coefficient budget. Across eight historically exposed, TRACE-selected protocols, this rule improves over matched confidence weighting, including 3.31 mIoU on COCO Object and 9.57 on the local COCO cat/dog protocol. Local cat/dog and Waterbirds mask-construction histories are not fully recoverable, limiting independent reconstruction. Falsifiable mechanism tests qualify the gains. Independent candidate-risk information is not established; spatial specificity is supported on Object but unresolved on PC60; and ranking effects depend on the weight distribution. A single probe loses to declared legacy components on Object and wins on PC60. Transfer to the declared SCLIP implementation is negative. Five fixed corruption curves favor TRACE at severities 1 and 3, while four severity-5 intervals cross zero. Finally, neither dataset has an eligible near-time match in the fixed compute grid, so equal-time benefit remains unestablished. TRACE is therefore an operational allocation rule with conditional empirical benefits, not a calibrated safety score or universal robustness guarantee.

1 Introduction

Open-vocabulary semantic segmentation (OVSS) uses text to specify dense recognition categories, extending vision–language representations beyond a closed training label set (Radford et al. 2021; Hajimiri, Ben Ayed, and Dolz 2025). Test-time adaptation (TTA) adjusts the predictor using unlabeled inputs. Entropy objectives can sharpen both correct and incorrect predictions (Wang et al. 2021; Noori et al. 2025); consequently, choosing how predictions contribute to an update matters as well as choosing the update objective.

A contextualized prediction supplies confidence but not a direct measurement of its response to changed surroundings. A token inside a predicted region can encode information from elsewhere in the image. Context can help recognition (Xiao et al. 2021); removing it from every adaptation input is therefore not our objective. Instead, we ask whether an additional, candidate-preserving observation can be used to distribute self-supervision while adaptation and final inference retain the original image.

TRACE implements this distinction with one fixed context probe per candidate. Connected components of the reset model's prediction define regions; initial confidence selects a bounded top-K pool. For each selected region, the probe preserves its pixels, changes only the complement, and measures its original text-query probability at its original token locations. The response supplies a detached weight, not a new label. Averaging entropy within each region and normalizing weights within each source image turns this response into relative allocation rather than an explicit region-area or cross-image support multiplier.

Matched results support the usefulness of this operational rule, but do not identify an independent risk estimator. Under eight fixed, historically exposed configurations, TRACE improves over initial-confidence weighting; the largest differences occur on COCO-CD and Waterbirds-Seg, with the latter explicitly containing development overlap. Subsequent tests expose important boundaries: matching candidate quality does not establish independent risk information, spatial contrasts are dataset-dependent, and crossed ranking/distribution changes have conditional effects. These results separate an observed allocation benefit from a claim about why every candidate should receive its weight.

The same distinction is necessary for simplification, transfer, and cost. A single context probe is not uniformly more accurate than the declared legacy diagnostic components. The declared SCLIP transfer is negative, and strong-corruption intervals are mixed. Moreover, the completed fixed accuracy–time grid supplies no eligible near-time comparator on either dataset. These are limitations of the empirical claims, not reasons to select new operating points after observing outcomes.

Our contributions are a candidate-preserving context response, an image-balanced regional entropy objective, and fixed-candidate tests of their conditional behavior. Crossed mechanism, legacy, transfer, severity, and compute studies distinguish an allocation benefit from unsupported risk,

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universal-transfer, and equal-time interpretations without post-result tuning.

2 Related Work

Dense vision–language prediction. OVSS methods transfer image–text representations through dense matching, mask classification, or segmentation-oriented adaptation (Ghiasi et al. 2022; Liang et al. 2023; Cho et al. 2024). Patch alignment and local self-distillation address the gap between global pretraining and spatial features (Mukhoti et al. 2023; Wu et al. 2024). NACLIP modifies spatial-token interactions, while SCLIP introduces correlative self-attention for training-free dense prediction (Hajimiri, Ben Ayed, and Dolz 2025; Wang, Mei, and Yuille 2024). TRACE acts after this inference interface has produced candidates: its question is how their unlabeled entropy terms share an adaptation objective.

Test-time self-supervision. TENT adapts normalization-affine parameters by minimizing entropy (Wang et al. 2021). CoTTA, EATA, and SAR study consistency, sample selection, and optimization control (Wang et al. 2022; Niu et al. 2022, 2023). TPT and WATT adapt vision–language models through prompts or template-associated weights (Shu et al. 2022; Osowiechi et al. 2024). MLMP combines visual levels and prompt-specific entropy for OVSS adaptation (Noori et al. 2025). TRACE uses prompt-specific dense entropy but allocates its regional contributions using a separate context response. Reported external methods are reference operating points, not independently reproduced or equally tuned controls in this study.

Context probing and interpretation. Background-dependence studies distinguish observed recognition from the evidence sustaining it (Xiao et al. 2021); perturbation-based explanations inspect model responses to controlled input changes (Fong, Patrick, and Vedaldi 2019). TRACE uses a specified perturbation to obtain update weights, rather than training an explanation model. Preserving pixels does not remove contextual feature interactions, shared errors, or confounding by initial candidate quality. Accordingly, the method does not treat its response as a calibrated correctness probability or an identified causal effect.

3 Method

Each episode starts from model parameters θ_0 and a reset optimizer. TRACE constructs a fixed candidate pool, measures one context response per candidate without gradients, and adapts visual LayerNorm affine parameters on the unmodified episode. Text embeddings and other model parameters remain frozen. Figure 1 summarizes the data flow; annotations never enter candidate construction, probing, or optimization.

3.1 Fixed Candidate Regions

A source image may supply multiple encoder crops, but remains the unit of objective normalization. Let b index crops and $j(b)$ their source images. The reset diagnostic readout defines

$$\pi^0_{b,i}(c \mid x_b) = \mathrm{softmax}_{c\in\mathcal{C}}\, z^{\mathrm{eval}}_{\theta_0}(x_b)_{i,c}, \tag{1}$$

where i indexes spatial tokens and $\mathcal{C}$ is the text-query vocabulary. Query extensions can make $\mathcal{C}$ differ from the evaluation-category set. Diagnosis uses entropy-weighted visual-layer aggregation and normalized mean prompt embeddings; optimization instead uses separate prompt distributions. Supplement A.3 gives the exact readouts.

Four-neighbor connected components of the initial query argmax define candidates $R = (b_R, \Omega_R, c_R)$. The region records its crop, token set, and original predicted query. Its initial confidence is

$$\kappa_R = \frac{1}{|\Omega_R|} \sum_{i\in\Omega_R} \pi^0_{b_R,i}(c_R \mid x_{b_R}). \tag{2}$$

Proposals from all crops of source image j compete for its top-K positions by descending confidence, with deterministic ties. Selected regions form $\mathcal{R}_j$. Components are neither stitched across crops nor merged across query extensions. Confidence bounds diagnostic workload, but does not certify candidate quality or establish confidence-independent admission.

3.2 Candidate-Preserving Context Response

The context probe changes a candidate's surroundings while holding its input pixels and readout locations fixed. Denormalize its crop to RGB pixels u, and upsample the original token mask to M_R by nearest-neighbor interpolation. With full-crop channel means μ, define

$$A(u) = \mathrm{clip}_{[0,255]}\big(\mathrm{Shift}(\mu + 0.35(u-\mu))\big), \tag{3}$$

$$x^{\mathrm{ctx}}_R = \mathrm{Norm}\big(M_R \odot u + (1-M_R)\odot A(u)\big), \tag{4}$$

where $\mathrm{Shift}(r,g,b) = (g,b,r)$. TRACE measures

$$\rho_R = \frac{1}{|\Omega_R|} \sum_{i\in\Omega_R} \pi^0_{b_R,i}(c_R \mid x^{\mathrm{ctx}}_R). \tag{5}$$

No new region is extracted from the transformed prediction. One diagnostic input is formed per selected candidate; bounded batching changes encoder-call count, not input count. Scores are detached and remain fixed throughout the episode.

The measured probability is support under this particular context transformation, not invariance to arbitrary contexts. Full-crop means and contextual feature interactions remain, and the original prediction can be wrong in both views. TRACE therefore uses ρ_R operationally without assuming it predicts unknown ground-truth update utility.

3.3 Image-Balanced Regional Entropy

The objective separates regional aggregation, within-image priority, and source-image budget. For prompt $a \in \{1,\dots,J\}$, let $p^a_{\theta,b,i}$ be the current query distribution on an unmodified crop. With $H(p) = -\sum_c p(c)\log p(c)$, regional entropy is

$$\overline{H}_R(\theta) = \frac{1}{J|\Omega_R|} \sum_{a=1}^{J} \sum_{i\in\Omega_R} H(p^a_{\theta,b_R,i}). \tag{6}$$

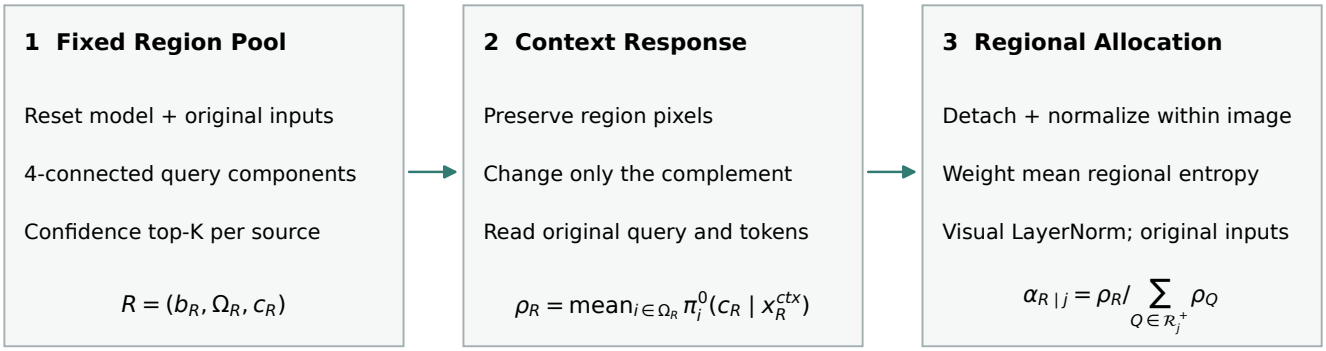


Figure 1: The fixed TRACE pipeline. Reset predictions define confidence-selected connected regions. Each diagnostic input preserves one region and changes its surroundings; the original query is read at the original locations. Detached supports normalize within source images and weight mean regional prompt entropy on unmodified inputs. The final segmentation is reconstructed from adapted query logits. Support is an allocation measurement, not a safety certificate.

Active regions satisfy $\rho_R > \epsilon$, with fixed $\epsilon = 10^{-8}$; write their set as $\mathcal{R}_j^+$ and the active source images as $\mathcal{I}_+$. The loss is

$$\alpha_{R|j} = \frac{\text{sg}(\rho_R)}{\sum_{Q \in \mathcal{R}_j^+} \text{sg}(\rho_Q)}, \tag{7}$$

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{I}_+|} \sum_{j \in \mathcal{I}_+} \sum_{R \in \mathcal{R}_j^+} \alpha_{R|j} \overline{H}_R(\theta). \tag{8}$$

If no image is active, adaptation is skipped. The candidate query selects the diagnostic response, not a cross-entropy pseudo-label. Each active image has total coefficient mass $1/|\mathcal{I}_+|$; region averaging removes an explicit area multiplier. Neither property equalizes gradient norms or Adam updates.

Relative allocation has algebraic limits. Common positive scaling of one image's scores cancels while its active set stays fixed, and a singleton candidate's positive score cancels entirely. Threshold crossings can still change support, and joint optimization couples regional gradients through shared parameters. These facts motivate the spatial and factorial tests below; they rule out interpreting absolute support as an image-level safety weight. Final inference reconstructs query logits on the original inputs, then takes the maximum over extensions of each evaluation category before category argmax.

4 Evaluation Protocol

The evaluation distinguishes fixed-operating-point performance from mechanism, transfer, and compute claims. The original complete-split comparison covers VOC20/VOC21 (1,449 images each), PC59/PC60 (5,105 each), COCO Object/Stuff (5,000 each), COCO-CD (500), and Waterbirds-Seg (5,322). It uses NACLIP ViT-L/14, its last 18 visual outputs, seven prompts, and one 224×224 input per source. Uniform, Confidence, and TRACE share candidates, order, readouts, and optimization settings; their weights are respectively one, κ_R, and ρ_R. No Adapt is the same pipeline's zero-update readout, not a claim to reproduce a literature plain-NACLIP pipeline.

Selection and exposure. The common probe precedes per-dataset parameter selection. Fixed 256-image training-source panels select batch size, learning rate, steps, and K for TRACE; the controls reuse those settings without independent tuning. The search contains 455 completed configurations with an 80-per-dataset cap. Configurations are fixed before the complete-split scores, but historical development/test exposure remains. Waterbirds additionally has 256 calibration images inside its unchanged evaluation population. Consequently, these are not untouched tests or fair independent tuning comparisons. Supplement B retains selection rules and coverage.

Local derived populations. COCO-CD is the retained 500-source COCO cat/dog protocol with background/cat/dog labels, not an additional official COCO task. Waterbirds-Seg retains all 5,322 local segmentation pairs: all 2,458 declared training sources occur in that full development population, including the selected 256 calibration sources. These nested counts are not additive and do not assert CLIP pretraining leakage. Their historical local mask/subset construction is not fully recoverable; the consumed loader, labels, splits, and remaining reconstruction gaps are specified in Supplement A.6. Their large gains are conditional on those retained local populations, not independent public-benchmark replications.

New fixed studies. Spatial (M02), legacy-component/objective (M03), factorial (M04), and compute-allocation (V01) studies use the complete Object and PC60 populations at their fixed settings. The isolated candidate-risk panel (M01) has 128 sources per dataset and is not a complete-split accuracy experiment. Generalization includes NACLIP B/16 on three clean populations, the declared TRACE multi-layer SCLIP B/16 on VOC20, all 15 predeclared VOC20 severity-5 corruptions, and five fixed corruption types at severities 1 and 3. Corrupted

inputs construct their own candidates and supports, without clean-image guidance. SCLIP uses the declared last 9 of 12 layers, not a reproduced published whole pipeline.

Geometry and uncertainty. All mIoUs pool per-class intersections/unions before taking the class mean, rather than averaging image IoUs. Native Object/PC60 studies resize labels with PIL nearest; G and G03 use the original palette-label decoder and OpenCV nearest at 224. These paths are not silently interchanged. New uncertainty estimates use 10,000 paired bootstrap draws of whole original source episodes, sharing draws across a study's arms. Object/PC60 have 385/393 episodes; VOC severity curves have 725 batch-two episodes including a singleton tail. Intervals are pointwise 95% linear percentiles without multiplicity correction. Undefined draws are retained and force a null interval. They are not independent-seed dispersion or uncertainty over tuning history. The original eight-protocol table remains deterministic seed-0 evidence without newly invented intervals.

Execution and outcome separation. Predictions are generated without annotation-content access and sealed before separate CPU scoring. Finite-value, numerical-agreement, and unchanged-label gates test implementation behavior, not semantic correctness or whole-pipeline equivalence. Paper tables project sealed summaries without new metric/bootstrap computation. Exact gate tolerances, failed attempts, and repaired fresh versions remain in the supplementary record.

5 Results

5.1 Fixed-Candidate Allocation

Context-probed weights improve over matched confidence weights at all eight selected operating points (Table 1). Object gains 3.31 mIoU, VOC21 3.13, and COCO-CD 9.57; PC60 and Stuff gain 0.21 and 0.11. Uniform itself falls below No Adapt on Object and VOC21, illustrating that selecting regions alone does not guarantee a beneficial update. These are conditional differences after replacing the weighting rule, including any EPS-active support effects, not proof of confidence-independent candidate utility.

Waterbirds reaches 56.08 mIoU versus 44.36 for Confidence and 47.46 for No Adapt. Its large difference is retained with the explicit development-overlap qualification rather than presented as held-out generalization. Published MLMP values and historical local references are separately attributed in Supplement I; different checkpoints, readouts, budgets, and selection histories prevent equal-protocol superiority claims.

5.2 What Does the Probe Identify?

Candidate risk is unresolved. M01 isolates 2,042 confidence-top-K candidates, resets the model/optimizer/RNG for each candidate, and applies a unit-weight update. Harm denotes a drop exceeding 0.001 in true-class support within the original region. At the final checkpoint, unadjusted high-minus-low-C harm-rate differences are -11.707 pp on Object and -28.431 on PC60. Conditioning on initial confidence leaves negative point differences, but additionally matching initial ground-truth quality reverses them to $+1.667$ and $+4.881$ pp. All 10,000 draws for these matched estimates have an undefined required denominator, so their intervals are null. Sparse coverage, confounding, and sign reversal do not establish independent risk information; they also do not establish that high C causes harm. Ground-truth quality is an offline diagnostic only, never a deployed score.

Spatial specificity is conditional. M02 retains the original readout/loss locations while comparing native region preservation, translated preservation, and a global context transform. Object native-minus-translated and native-minus-global differences are $+0.828$ and $+1.027$ pp, with positive intervals (Table 2). Both PC60 spatial intervals cross zero. Thus Object supports the declared preserved-position mechanism at this operating point, whereas PC60 does not resolve spatial specificity. Candidate-local causal effects are not identified, and the natural-active comparison can include support changes at EPS.

Ordering and distribution interact. M04 assigns the sorted Confidence or C weight values using Confidence or C ordering. On Object, C ordering helps with C distribution but hurts with Confidence distribution. On PC60, substituting C distribution has an interval crossing zero under C ordering and a negative interval under Confidence ordering. The difference-in-differences interaction is positive on both datasets; the common-active variant retains this qualitative pattern. No universal ranking or magnitude advantage follows. Every arm measures C for this diagnostic comparison, including the Confidence/Confidence arm, so its cost cannot stand in for a zero-probe compute control.

One probe is not uniformly more accurate. M03 compares single C with five pinned historical role/view/score components embedded in the current role-free candidate, optimizer, and readout engine. Under the image objective, C minus legacy is -0.380 pp on Object and $+0.142$ on PC60; both intervals exclude zero in opposite directions. Episode-normalized comparisons have the same directions. Objective-by-diagnostic interactions cross zero, which does not prove equivalence. The result concerns these declared components, not the published whole legacy system, old optimizer, or old execution engine.

5.3 Transfer and Corruption Severity

The allocation signal does not transfer uniformly across dense predictors. NACLIP B/16 C-minus-Confidence intervals are positive on Object, PC60, and VOC20 (Table 3), while Object and PC60 C point estimates remain slightly below their zero-update controls. The declared SCLIP B/16 VOC20 condition instead loses 2.395 pp with interval $[-3.512, -1.345]$. A tested official-component gate does not establish whole-pipeline equivalence; neither does the negative result license changing the declared readout or retuning to obtain positive transfer.

The 15-type severity-5 primary study is mixed rather than uniformly robust. Its fixed macro C-minus-Confidence difference is $+0.581$ pp with interval $[0.221, 0.947]$, with 11

Table 1: Full-population mIoU (%, seed 0). Uniform and Confidence use the TRACE-selected configuration within each dataset, without separate tuning. All splits retain historical development exposure; Waterbirds includes all 256 calibration sources among its 5,322 images. Δ is TRACE minus Confidence in percentage points; Object/Stuff denote COCO Object/COCO-Stuff.

Dataset	Images	No Adapt	Uniform	Confidence	TRACE	Δ
VOC20	1,449	79.04	81.93	82.24	83.61	+1.37
VOC21	1,449	49.66	47.36	50.25	53.38	+3.13
PC59	5,105	31.59	32.92	33.17	33.59	+0.41
PC60	5,105	28.95	30.59	30.57	30.78	+0.21
Object	5,000	29.65	27.51	29.36	32.67	+3.31
Stuff	5,000	20.19	21.80	21.80	21.91	+0.11
COCO-CD	500	62.82	60.15	64.66	74.23	+9.57
Waterbirds	5,322	47.46	41.62	44.36	56.08	+11.72

Table 2: Signed mechanism contrasts, mIoU percentage points with pointwise 95% intervals. M02 tests preserved-position spatial specificity; M03 compares declared diagnostics under the image objective; M04 crosses distribution and ordering in its natural-active study. Positive differences are not universal: Object reverses its ordering effect under Confidence distribution, and PC60's distribution effect is null or negative. Complete arm values, common-active tests, and all intervals are in the supplement.

Study / contrast	Object Δ [95% CI]	PC60 Δ [95% CI]
M02: Native - translated	+0.828 [+0.433, +1.236]	+0.105 [-0.029, +0.234]
M02: Native - global	+1.027 [+0.605, +1.446]	+0.117 [-0.015, +0.245]
M03: C image - legacy image	-0.380 [-0.742, -0.040]	+0.142 [+0.032, +0.250]
M04 natural: Order @ C dist.	+1.708 [+1.224, +2.152]	+0.641 [+0.483, +0.792]
M04 natural: Order @ Conf. dist.	-0.726 [-1.118, -0.308]	+0.102 [+0.057, +0.149]
M04 natural: Dist. @ C order	+4.036 [+3.284, +4.659]	+0.108 [-0.026, +0.240]
M04 natural: Dist. @ Conf. order	+1.602 [+0.985, +2.179]	-0.431 [-0.534, -0.330]
M04 natural: Dist.-order interaction	+2.433 [+1.845, +2.957]	+0.539 [+0.392, +0.683]

point wins and four losses. Noise and digital group intervals cross zero; elastic transformation has a negative interval. The full 15-condition table and its predeclared groups remain in Supplement E, separate from the five-type severity follow-up.

The new severity curves refine the claim to their fixed conditions (Figure 2). All five types at severities 1 and 3 favor C with positive pointwise intervals. At severity 5, motion blur remains positive, while Gaussian noise, fog, contrast, and JPEG intervals cross zero; Gaussian noise and contrast have negative point differences. The curve protocol was registered after the primary severity-5 results were known but before the new severity-1/3 predictions. Types and severities were fixed rather than selected from severity-5 gains. Reused severity-5 points receive the registered G03 bootstrap namespace, not an independent replication or an overwrite of the primary intervals. No monotonicity, slope, AUC, or new macro claim is inferred.

5.4 Accuracy Versus Actual Compute

V01 asks whether diagnostic compute outperforms extra Confidence updates, using a fixed grid rather than accuracy-selected stopping. Object uses C at 15/30 steps and Confidence at 30/60/120; PC60 uses C at 4/7 and Confidence at 7/14/28. Synchronized raw model time includes episode zero and the common finite-Adam observer, while decode, H2D, I/O, ground-truth scoring, auxiliary gates, and final cleanup are excluded and accounted separately. These measurements must not be mixed with historical warmup-excluded timings.

Neither dataset supplies an eligible near-time comparison (Figure 3). At the native C point, the nearest Confidence point minimizes absolute log time ratio, breaking ties by fewer steps. Object chooses Confidence-30 with ratio 0.826; PC60 chooses Confidence-14 with ratio 1.192. Both fail the fixed 10% tolerance, so both near-time accuracy outcomes are null. Same-step gains remain $+3.310$ pp on Object and $+0.210$ on PC60, but they do not answer the equal-time question. PC60's nearest-point difference of $+0.620$ pp is descriptive only. Longer Confidence runs in this grid do not justify retuning the grid or interpolating a matched winner.

5.5 Grouped Evidence and Fixed Cases

Source-level cached groups and class subsets describe heterogeneous outcomes, not candidate-local utility. Every Object source reorders; only two PC60 sources keep the same candidate ranking, and their interval is undefined in 1,315 of 10,000 draws. Identical ranking would not imply identical normalized values, active support, or joint gradients. Foreground/background and thing/stuff class-subset means use their declared class mappings and are not binary-role IoU. These cached results therefore do not establish a same-versus-reordered causal allocation benefit.

Pixel transitions likewise separate kinds of benefit: Object C-minus-Confidence correct retention rises 3.553 pp while wrong correction falls 4.815 pp. Net pixel correctness still improves 0.630 pp. This mixed event-level behavior is not a uniform correction gain or the pooled class-mIoU estimand.

Four fixed cases use original manifest indices 0 and 1 in Object and PC60, without selecting improvements. Supple-

Table 3: Declared B/16 clean transfer conditions. No Adapt, Confidence, and C share the same within-condition readout and settings. Values are pooled mIoU; C minus Confidence includes its paired pointwise 95% interval. The SCLIP row uses TRACE's declared multi-layer readout, not a published whole-pipeline reproduction.

Condition	NoAdapt	Confidence	C	C − Conf. [95% CI]
NACLIP B16 Object	29.977	27.943	29.952	+2.009 [+1.478, +2.535]
NACLIP B16 PC60	31.004	30.074	30.867	+0.793 [+0.630, +0.955]
NACLIP B16 VOC20	80.310	82.547	83.648	+1.100 [+0.334, +1.924]
SCLIP B16 VOC20	35.215	33.928	31.533	-2.395 [-3.512, -1.345]

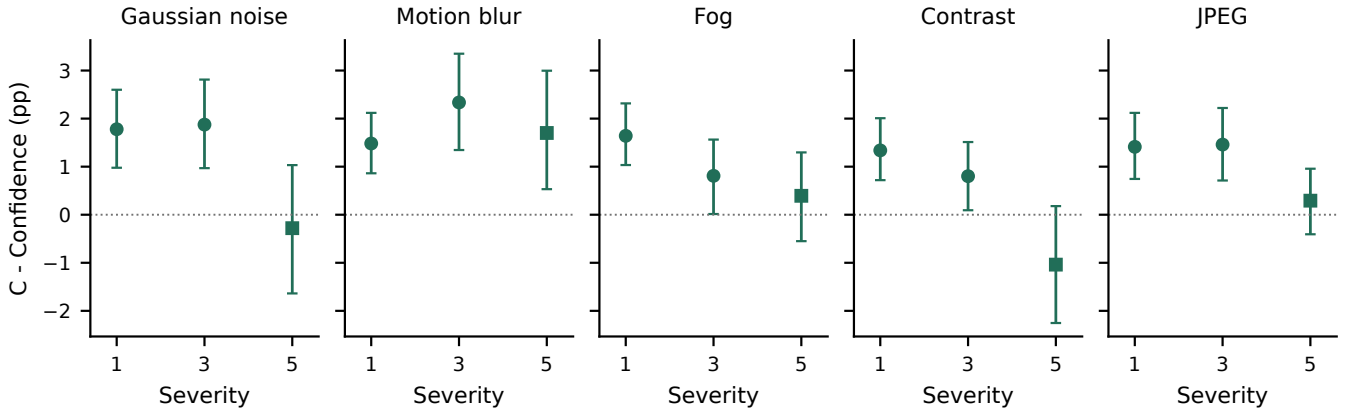


Figure 2: Five fixed corruption types at severities 1, 3, and 5: C minus Confidence, with pointwise 95% paired whole-episode intervals. Points and bars are sealed estimates, not a fitted trend or simultaneous confidence band. All ten severity-1/3 intervals are positive. At severity 5, only motion blur is positive; the other four intervals cross zero, including negative Gaussian-noise and contrast points. Severity-5 points reuse the original exact population/count evidence, not new inference.

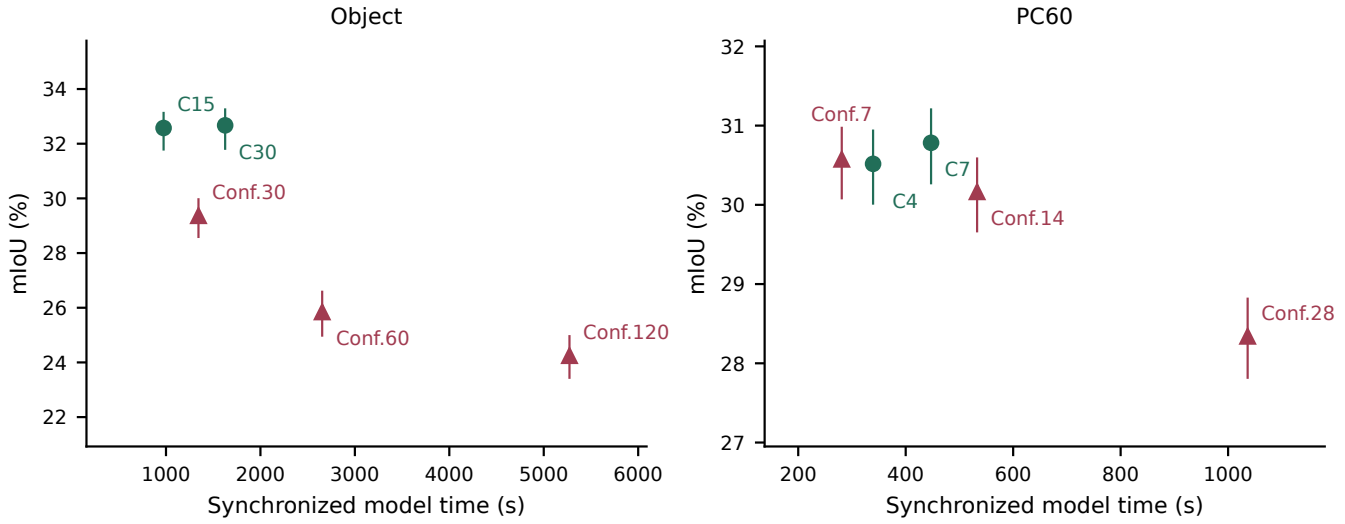


Figure 3: The complete fixed V01 accuracy–model-time grid. Each marker is an actual completed operating point; vertical bars are pointwise 95% whole-episode accuracy intervals. No interpolation, accuracy-based selection, or unexecuted point is shown. Step counts label the points. At native C, the time-only nearest Confidence/C ratios are 0.826 on Object and 1.192 on PC60, both outside $[0.9, 1.1]$. Consequently, both preregistered near-time accuracy comparisons are null, not equal-time wins.

ment H shows all four stored composites: input, measured support/overlay, ground truth, and actual saved readouts. One Object Confidence prediction is nearly background, but contains two palette colors rather than being strictly constant; it is retained. The green/magenta supports denote original/translated preservation locations, not attention or a dis-

played actual perturbation. These illustrations establish neither prevalence nor a causal effect of an individual candidate.

6 Discussion and Limitations

The strongest supported statement is conditional: a fixed context response can improve regional allocation under the declared configurations. This is weaker than independent update-risk calibration. Initial prediction quality remains a confounder, candidate admission is confidence-selected, active-set changes matter, and shared-parameter Adam interactions prevent a regional coefficient from being read as isolated utility. Object spatial and factorial tests constrain the mechanism, but PC60 spatial intervals and M01 matching do not establish it universally.

Simplification, transfer, and compute each have distinct limits. Fewer diagnostic inputs do not guarantee better accuracy, as Object's legacy comparison shows; a positive NACLIP comparison does not imply positive SCLIP transfer; and mild-corruption gains do not imply severe-corruption gains. One input per candidate reduces diagnostic input count relative to the legacy two/three-view schedule, not total wall time, and V01 leaves equal-time benefit unestablished. Null intervals are not evidence of equivalence, while pointwise intervals across many tests are not multiplicity-corrected discovery claims.

Historical exposure limits external validity despite prediction/scoring separation. Controls are not independently tuned; Waterbirds has development overlap; native/G geometries differ; and seed-0 runs do not estimate training-seed variability. Eight protocols are not eight independent domains: VOC20/21, PC59/60, and Object/Stuff reuse source families under different label protocols. The study does not establish arbitrary-model, new class-shift, candidate-causal, or whole-SCLIP/legacy-pipeline claims. These boundaries do not justify changing the fixed experiment after outcomes.

7 Conclusion

TRACE uses a candidate-preserving context response to distribute regional entropy within source images. Fixed-candidate comparisons show gains over confidence weighting, while the mechanism tests identify conditional interactions rather than independent risk. Opposite legacy results, negative SCLIP transfer, strong-severity nulls, and unmatched compute comparisons delimit the empirical claim. The resulting contribution is a specified, auditable allocation rule and its bounded evaluation, not a universal or equal-time guarantee.

AI Assistance

AI assistance was used for source drafting, experimental orchestration, mathematical consistency checks, visualizations, and table-provenance tooling. The authors remain responsible for the evidence and final text.

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